12

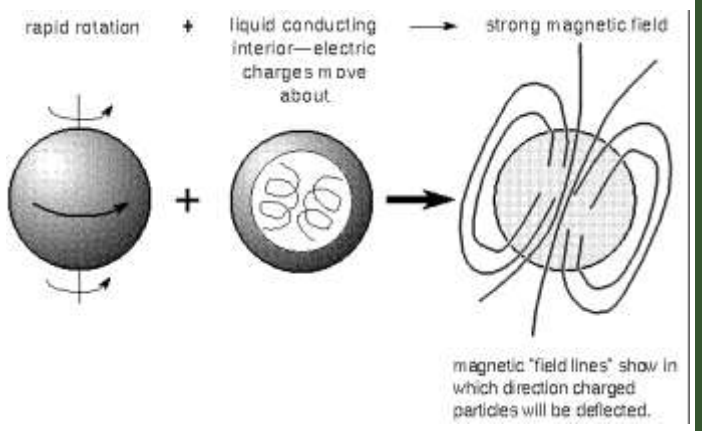
Magnetic Fields and Electromagnetism

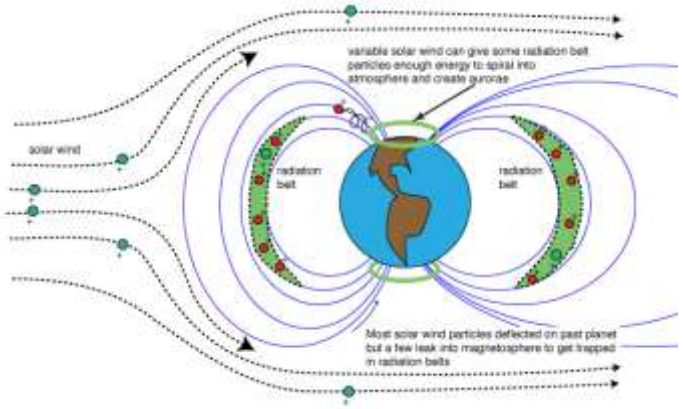
12.2

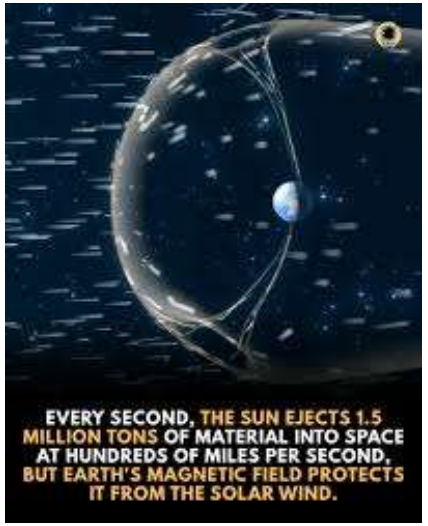
Magnetism and Moving Charges

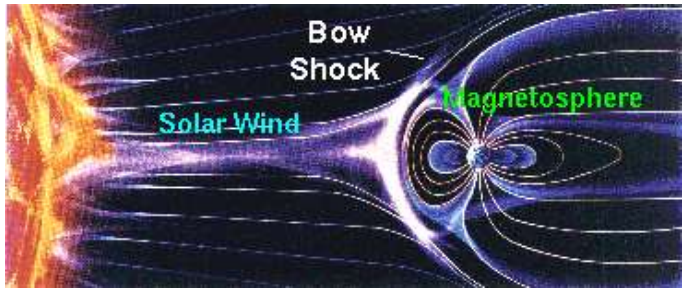
12.2.1

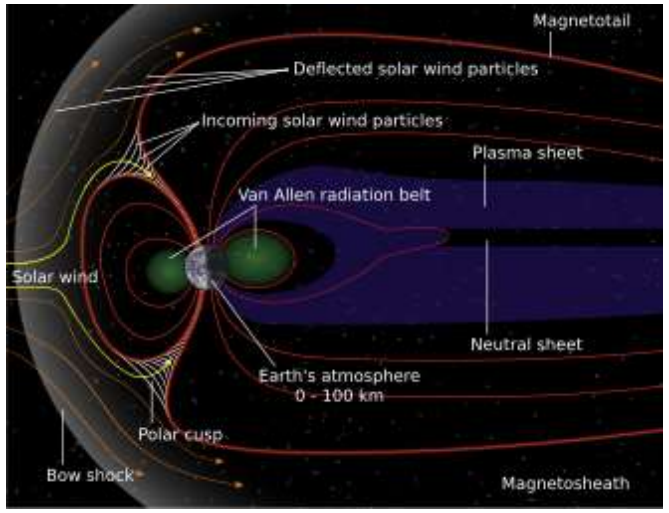
Moving Charges -> Magnetic Fields and Right Hand Rules to determine the direction of “B”

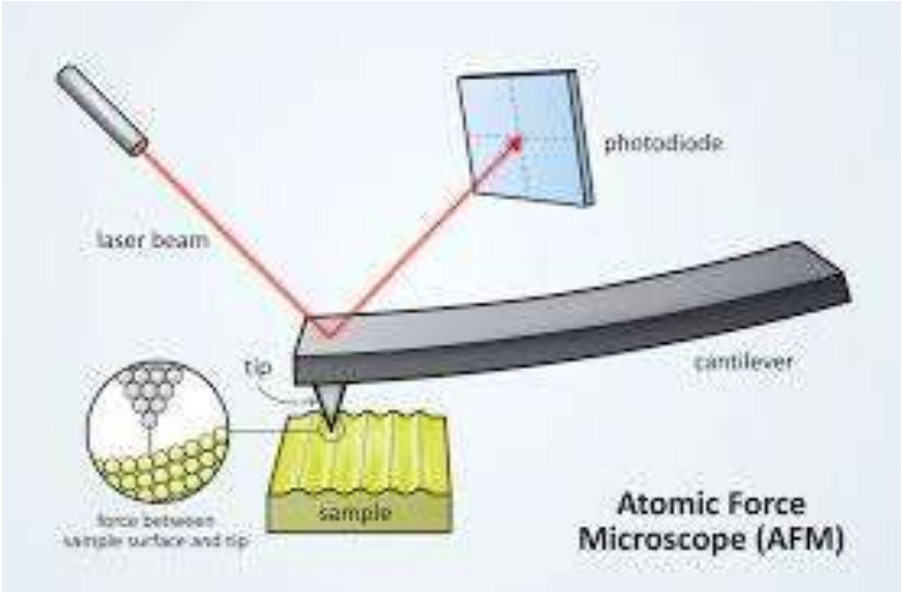

rapid rotation
+
liquid conducting interior—electric charges move about
strong magnetic field
magnetic "field lines" show in which direction charged particles will be deflected.

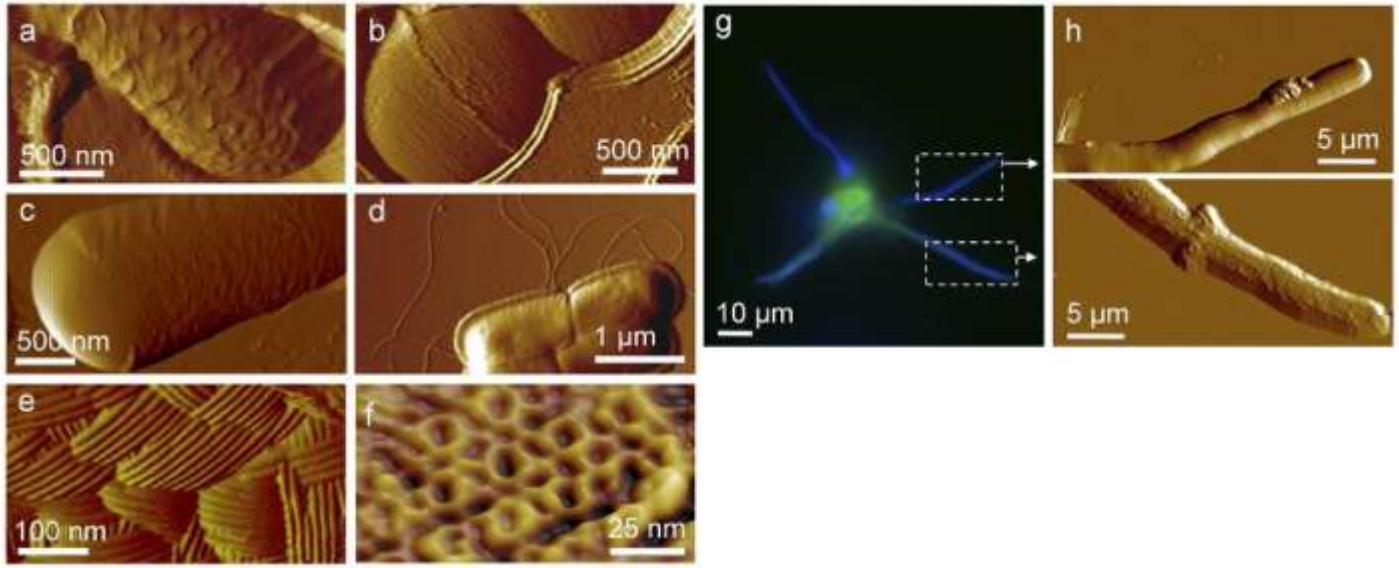

variable solar wind can give some radiation belt particles enough energy to spiral into atmosphere and create auroras
solar wind
radiation belt
radiation belt
Most solar wind particles deflected on past planet but a few leak into magnetosphere to get trapped in radiation belts

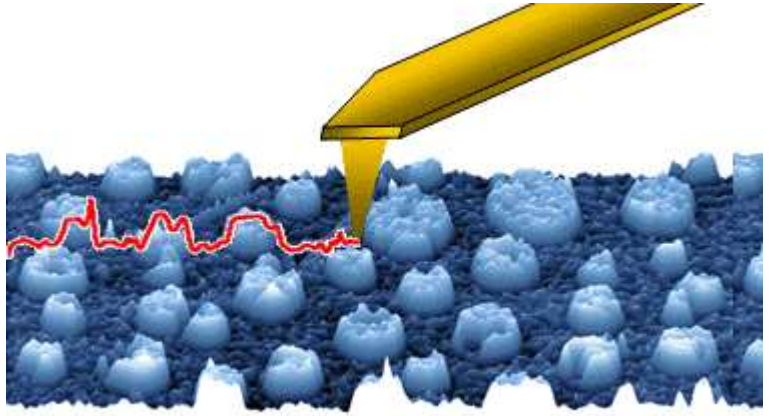

EVERY SECOND, THE SUN EJECTS 1.5 MILLION TONS OF MATERIAL INTO SPACE AT HUNDREDS OF MILES PER SECOND, BUT EARTH'S MAGNETIC FIELD PROTECTS IT FROM THE SOLAR WIND.

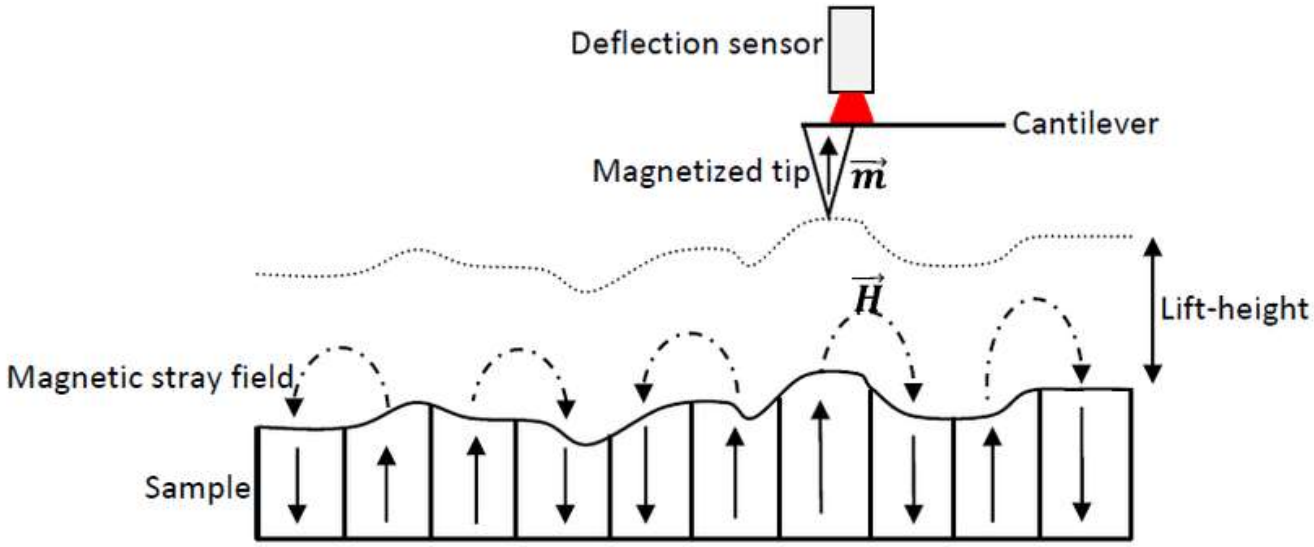

Bow Shock
Solar Wind
Magnetosphere

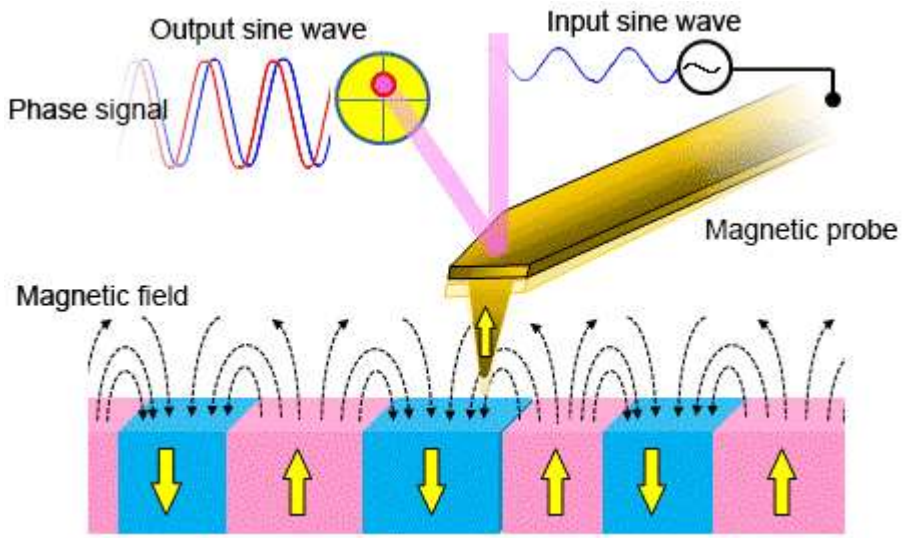

Magnetotail
Deflected solar wind particles
Incoming solar wind particles
Plasma sheet
Van Allen radiation belt
Solar wind
Neutral sheet
Earth's atmosphere
0 - 100 km
Polar cusp
Bow shock
Magnetosheath

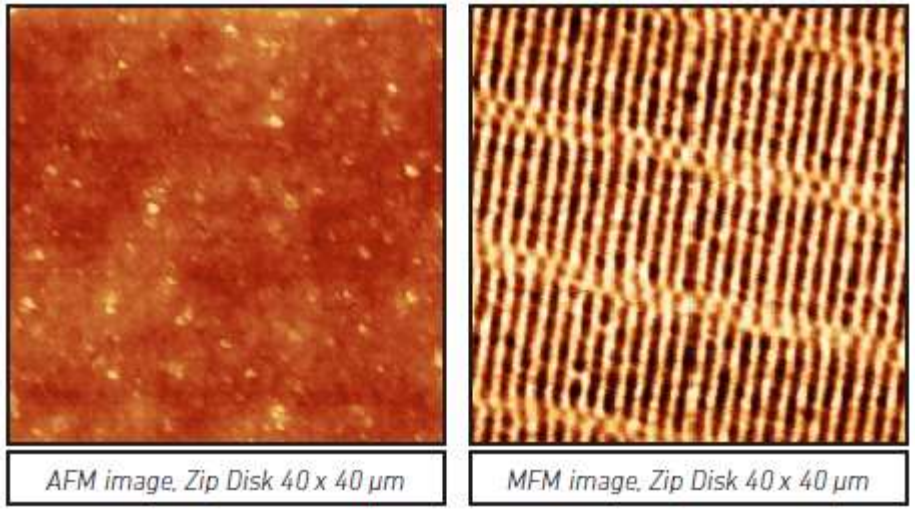

photodiode
laser beam
cantilever
tip
sample
force between
sample surface and tip
Atomic Force
Microscope (AFM)


a
500 nm
b
500 nm
c
500 nm
d
1 µm
e
100 nm
f
25 nm
g
10 µm
h
5 µm
5 µm



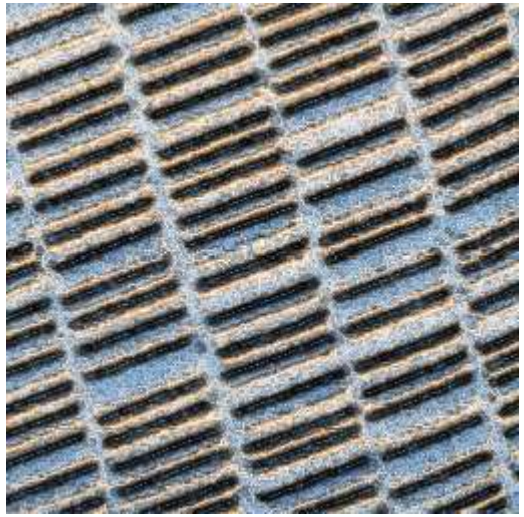


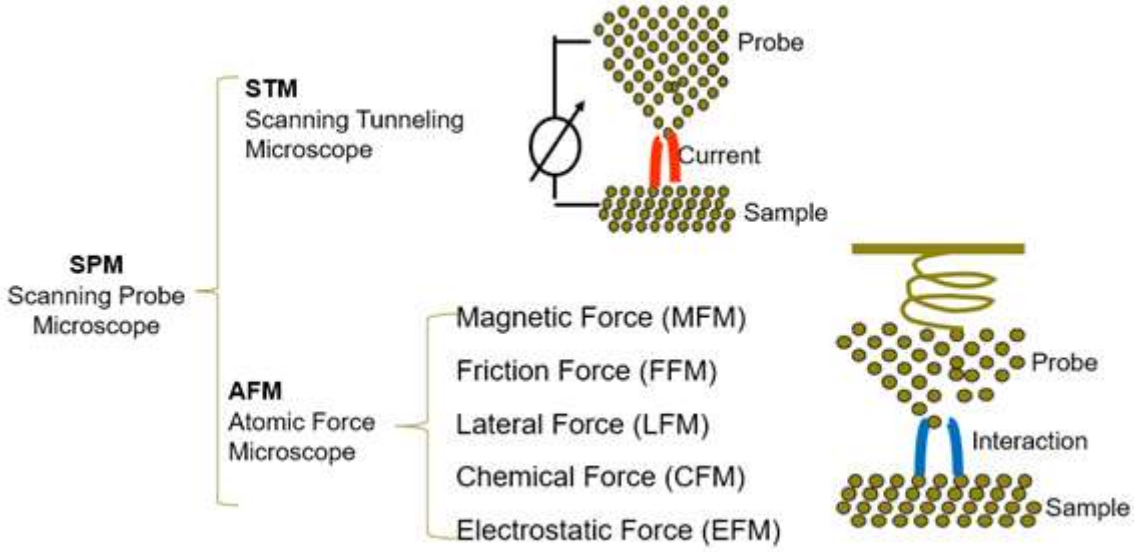


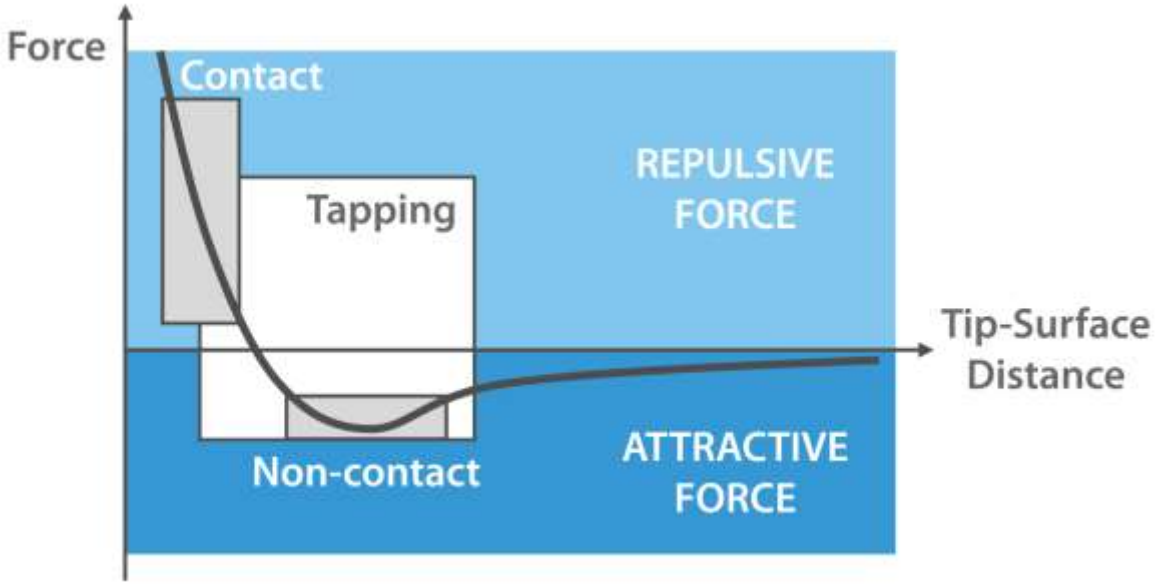


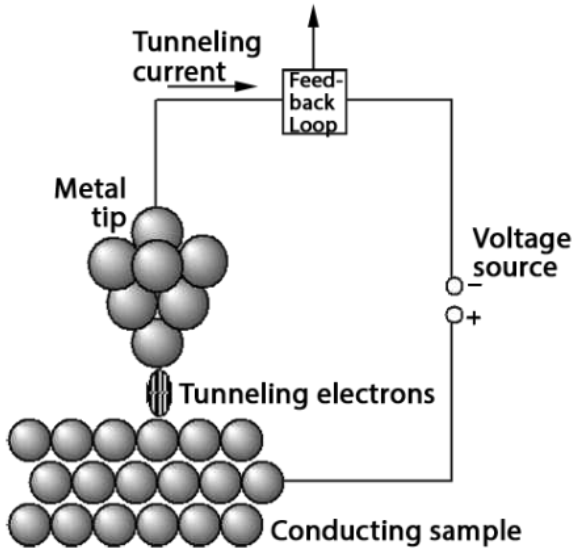
AFM image, Zip Disk 40 x 40 µm

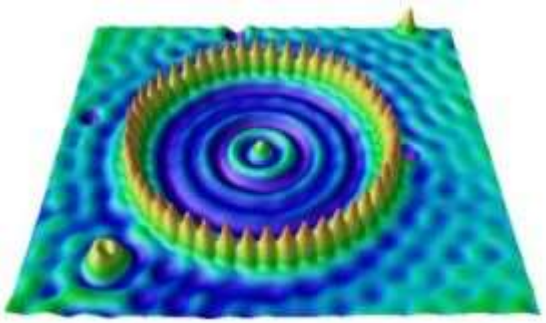
MFM image, Zip Disk 40 x 40 µm




SPM
Scanning Probe Microscope
STM
Scanning Tunneling Microscope
Probe
Current
Sample
AFM
Atomic Force Microscope
Magnetic Force (MFM)
Friction Force (FFM)
Lateral Force (LFM)
Chemical Force (CFM)
Electrostatic Force (EFM)
Probe
Interaction
Sample


Force
Contact
Tapping
REPULSIVE FORCE
Tip-Surface Distance
Non-contact
ATTRACTIVE FORCE


Tunneling current
Feed-back Loop
Metal tip
Voltage source
–
+
Tunneling electrons
Conducting sample



Magnetic Fields

Moving charge carriers produce magnetic fields.

The magnitude of a magnetic field is defined by the **Biot-Savart Law**:

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I d\vec{l} \times \hat{r}}{r^2}$$

Vector Product

AKA “Cross Product”

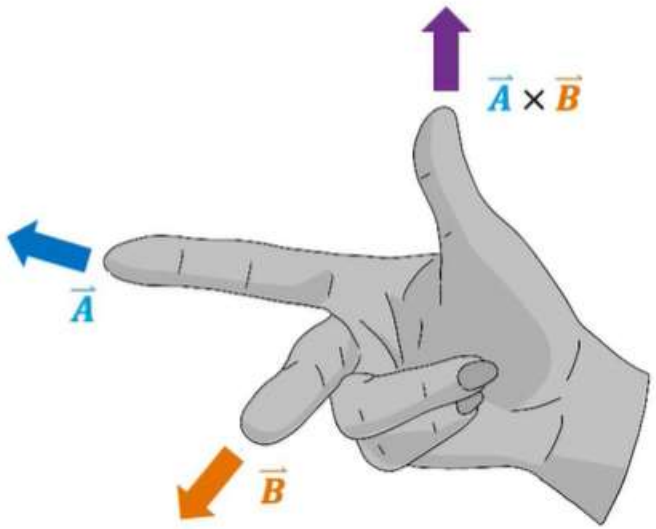
$$\vec{C} = \vec{A} \times \vec{B}$$

Vector $\vec{C}$ is perpendicular to vectors $\vec{A}$ and $\vec{B}$.

We find the direction of $\vec{C}$ by using a **right-hand rule**.

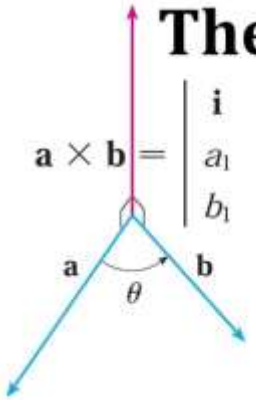
Your pointer and middle fingers keep this orientation.

Your thumb is kept perpendicular to those fingers.



The thumb is “mutually orthogonal” to both other fingers

The Cross Product



$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = (a_2b_3 - a_3b_2)\,\mathbf{i} - (a_1b_3 - a_3b_1)\,\mathbf{j} + (a_1b_2 - a_2b_1)\,\mathbf{k}$$

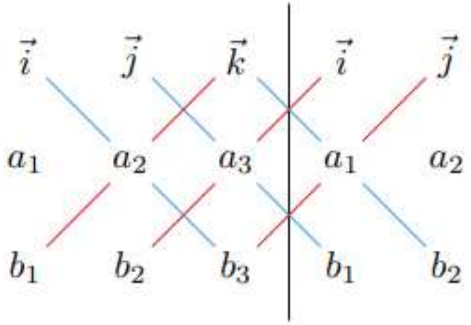
Determinant Form

Matrix Representation of Dot Product

cuemath

$$\vec{A}^T = \begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix} \qquad \vec{B} = \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix}$$

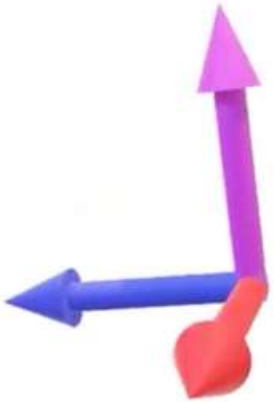
$$\begin{bmatrix} A_1 & A_2 & A_3 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ B_3 \end{bmatrix} = A_1B_1 + A_2B_2 + A_3B_3 = \vec{A}.\vec{B}$$



$$\vec{a} \times \vec{b} = \vec{i}(a_2b_3 - a_3b_2) + \vec{j}(a_3b_1 - a_1b_3) + \vec{k}(a_1b_2 - a_2b_1)$$

$$\theta = \sin^{-1}\left[\frac{|\vec{a} \times \vec{b}|}{|\vec{a}|\,|\vec{b}|}\right]$$

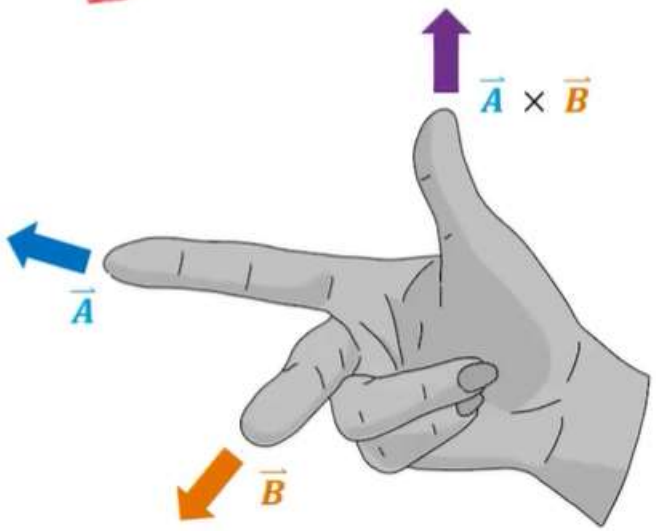
Right-Hand Rule



Always use your right hand.

Keep your fingers in the same position and move your whole hand to get the correct alignment.

Never swap the positions of your pointer finger and your middle finger.



Magnetic Fields

$I d\vec{l}$ is in the **direction** of the current

μ_0 is the permeability of free space.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

$\hat{r}$ is the **direction** of the radius.

dB is in the **direction** of the magnetic field

r is the distance from the charged object to the point where the field is measured.

Magnetic Fields

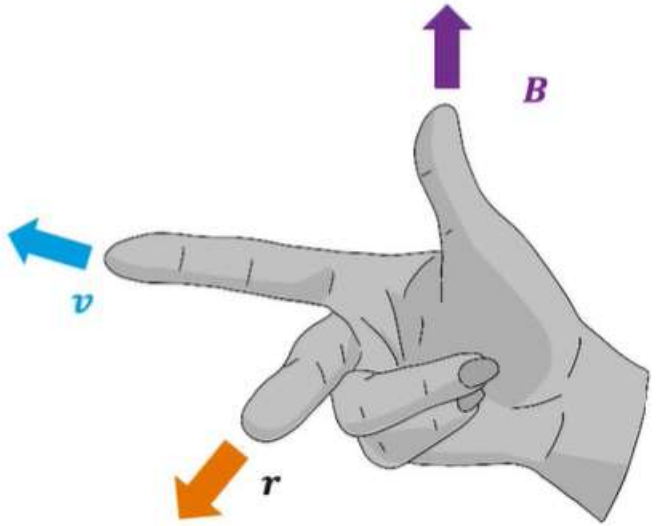
$\hat{r}$ is the **direction** of the radius.

v is in the **direction** of the moving charge

$$B = v \times \hat{r}$$

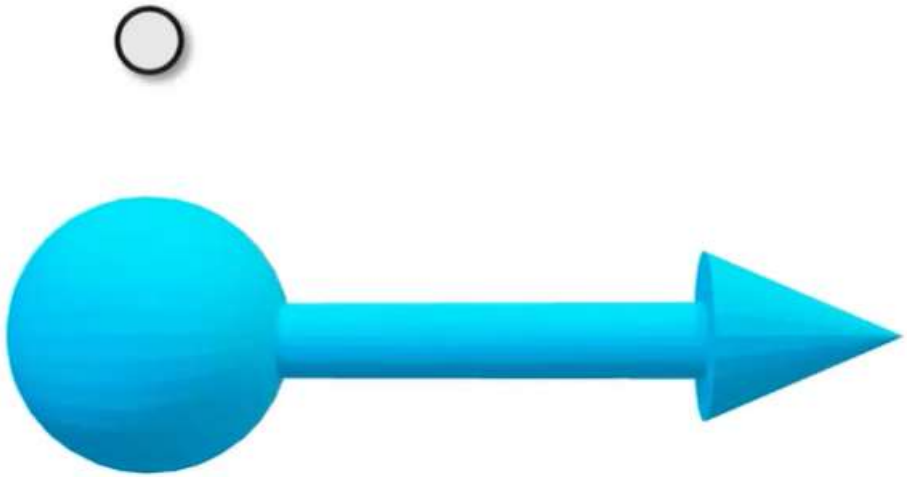
*This is an equation of direction only and is not used to calculate value.

B is in the **direction** of the magnetic field



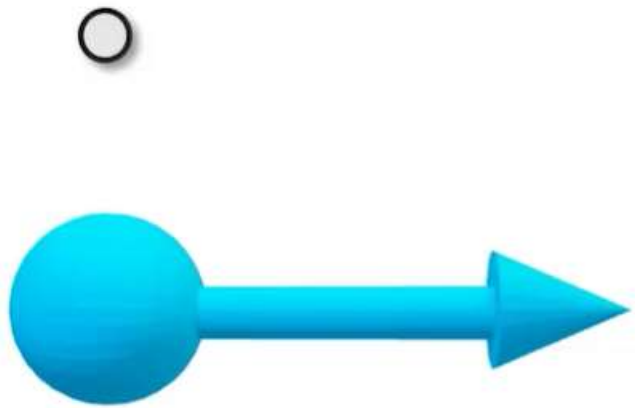
Moving Positive Charge Carrier

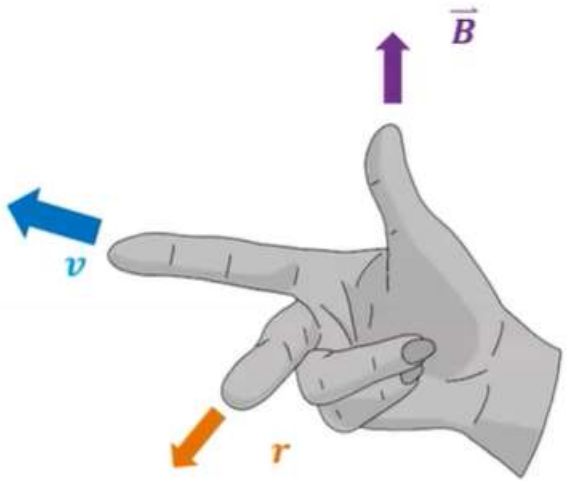
A positively charged object is moving with a constant speed of v in the direction shown. What is the direction of the magnetic field at a point directly above the charged object?



Moving Positive Charge Carrier

Use the right-hand rule:

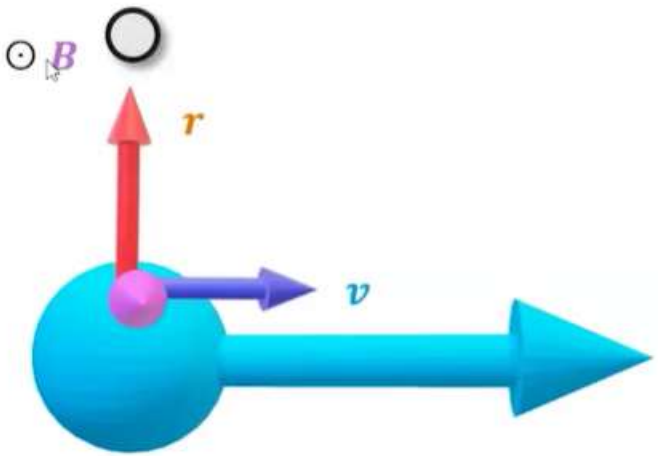




The **velocity** (your pointer finger) is to the right.

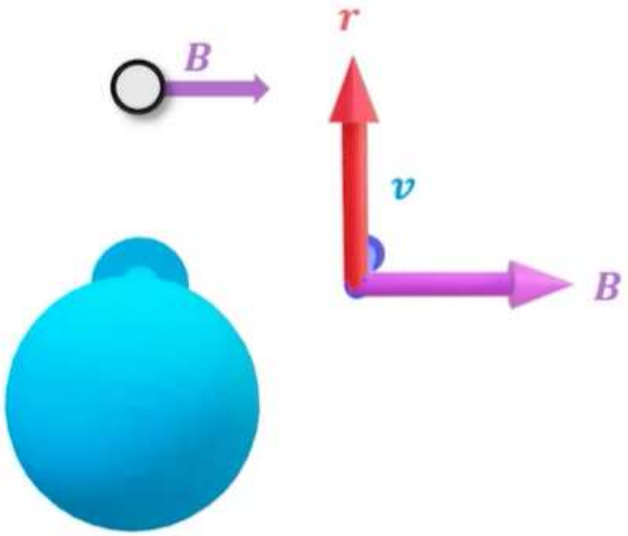
The **radius** (your middle finger) is up.

Moving Positive Charge Carrier



The **magnetic field** is directed out of the page at the chosen point.

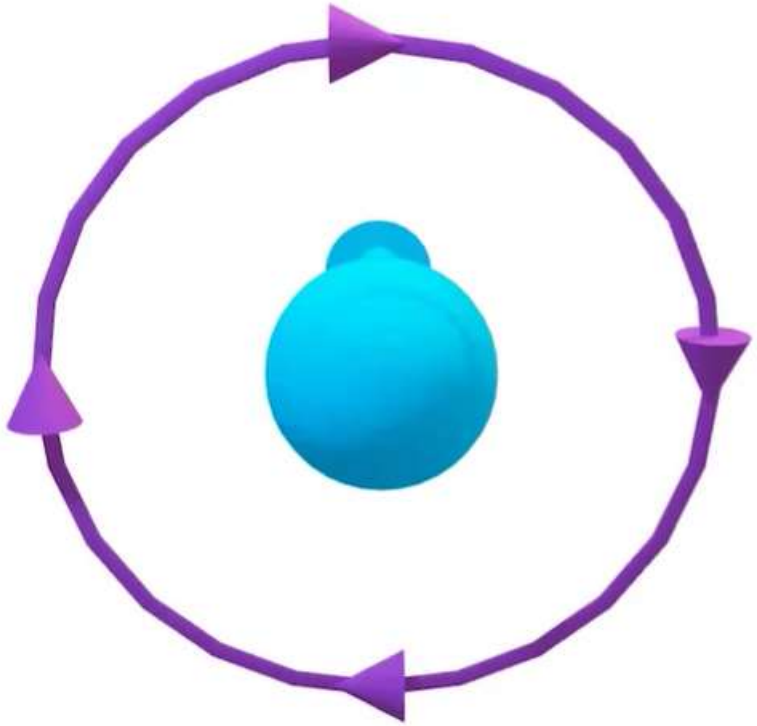
Moving Positive Charge Carrier



In this view of the same scenario, the velocity is directed into the page.

If we continue finding the direction of the field at various points the same distance from the charged object, we see that the field forms a circle around the object.

Moving Positive Charge Carrier



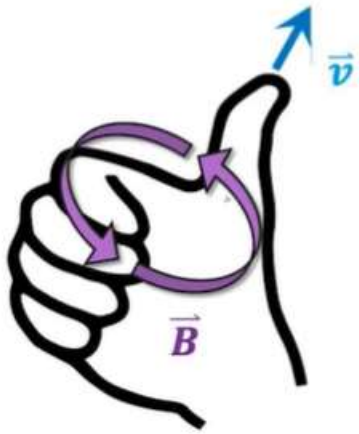
In this view of the same scenario, the velocity is directed into the page.

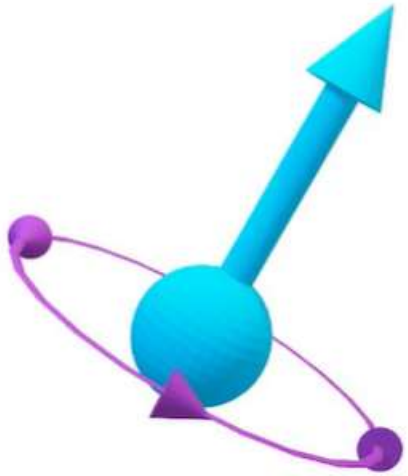
If we continue finding the direction of the field at various points the same distance from the charged object, we see that the field forms a circle around the object.

Another Right-Hand Rule

Point your thumb in the direction of the object's velocity.

Your fingers curl in the direction that the field circles around the moving charged object.





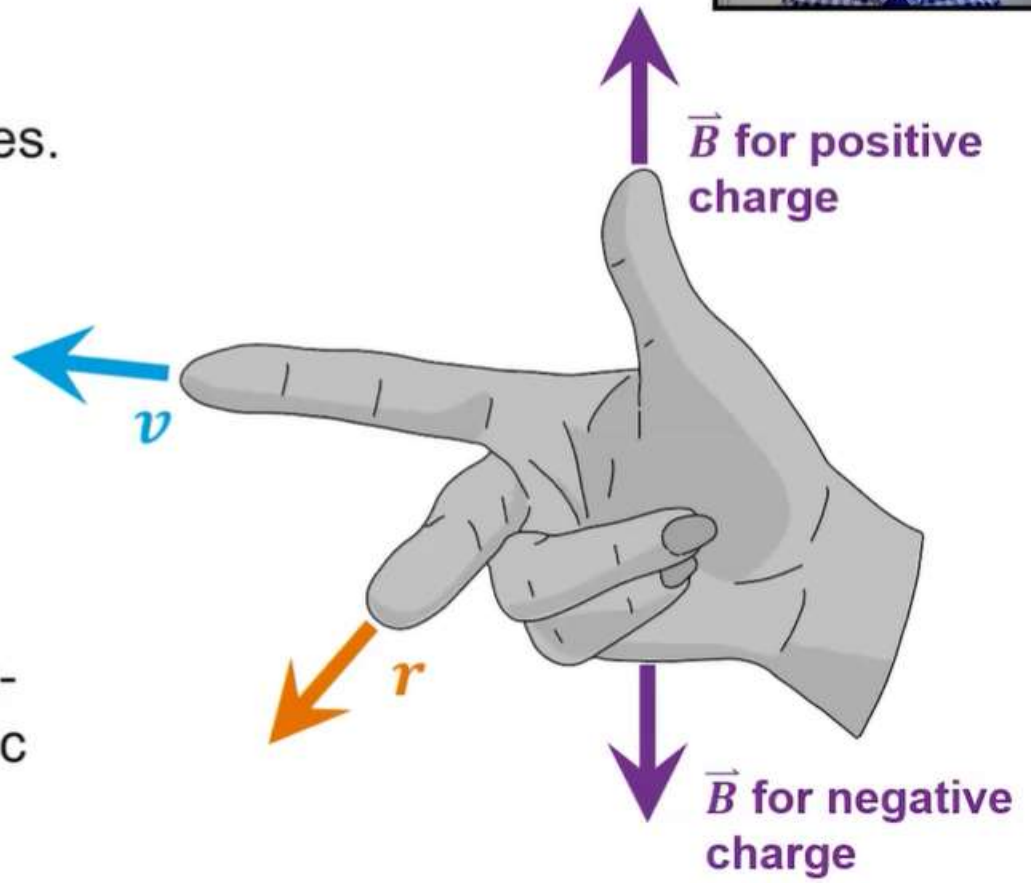
Negatively Charged Objects

AKA Electrons!

Negatively charged objects have the opposite sign compared to positive charges.

The magnetic field created by a moving negatively charged object will have the opposite direction of the magnetic field created by a positively charged object.

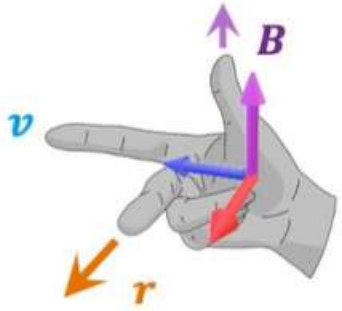
If you need the direction of the magnetic field from a negative charge, use the right-hand rule, but remember that the magnetic field will point in the opposite direction of your thumb.



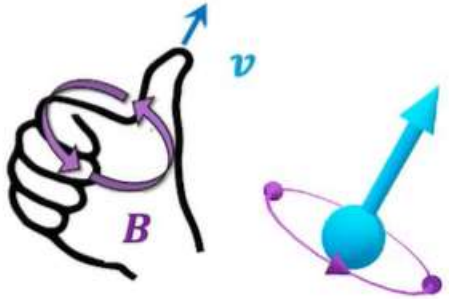
Takeaways

Moving charged objects produce magnetic fields.

The direction of the magnetic field created by a moving positive charge is found using the right-hand rule.



The direction of the magnetic field created by a moving negative charge has the opposite direction of the field created by a moving positive charge.



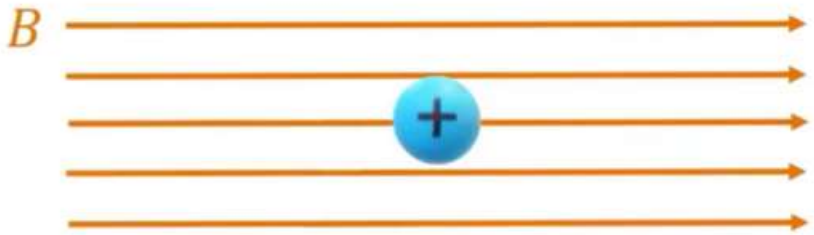
12.2.2

Force on a Charged Particle moving through a Magnetic Field

Right Hand Rule “#3” to determine the direction of the Force

A Charged Object at Rest

This is a uniform magnetic field directed to the right. It has a magnitude of B.

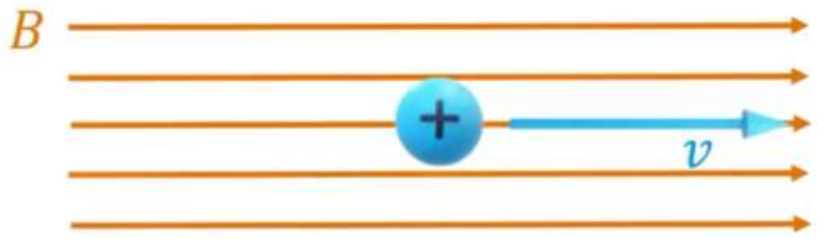


An object with positive charge is placed in the magnetic field.

A magnetic field does not exert a force on a charged object at rest.

Parallel Velocity

The positively charged object is given a velocity in the same direction as the magnetic field.

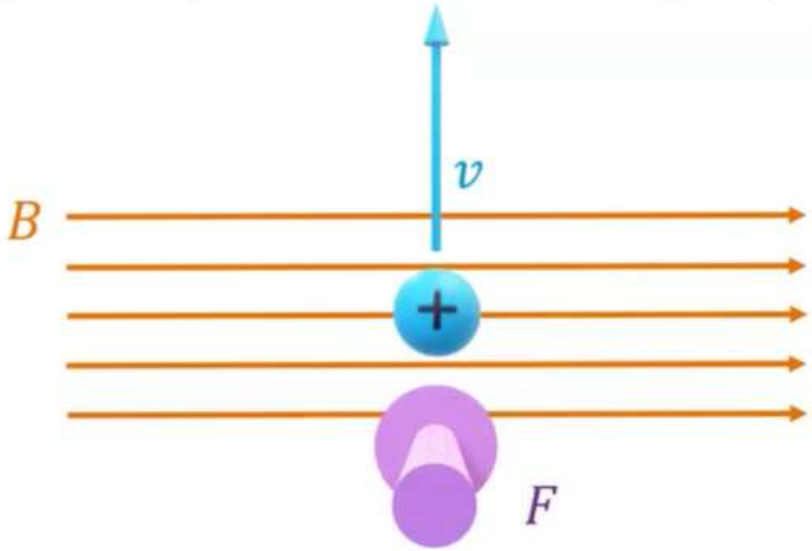


It still experiences no force from the magnetic field.

Not even if it goes in the opposite direction.

Perpendicular Velocity

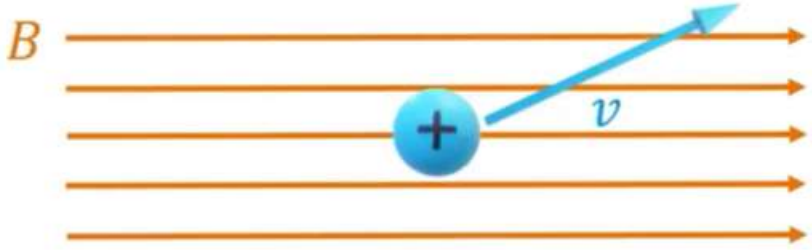
The positively charged object now moves perpendicular to the magnetic field.



The magnetic field exerts a force on the charged object that is directed into the page, perpendicular to both the velocity and the magnetic field.

Velocity at an Angle

The positively charged object is given a velocity that is at some non-perpendicular angle to the field

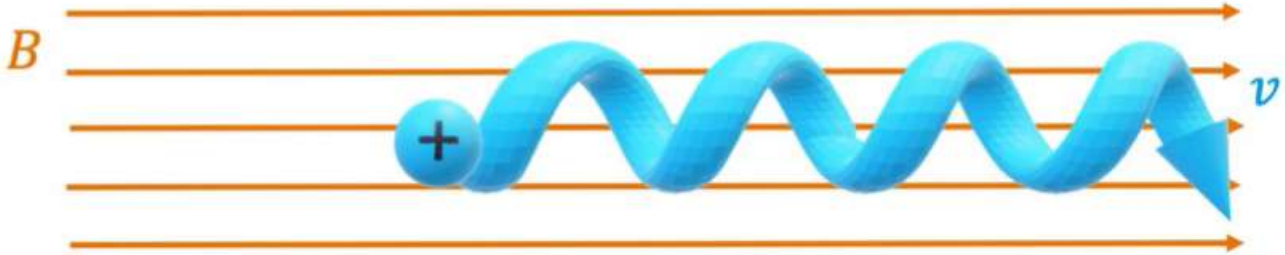


The magnetic field will exert a force on the particle into the page, but this force will only affect the component of velocity that is perpendicular to the field.

The parallel component will remain unchanged.

Velocity at an Angle

The particle will have two components of motion. The parallel, unchanged component will be in the direction of the magnetic field. The perpendicular component will experience a force that is always perpendicular to its motion, causing it to move in a circular pattern.



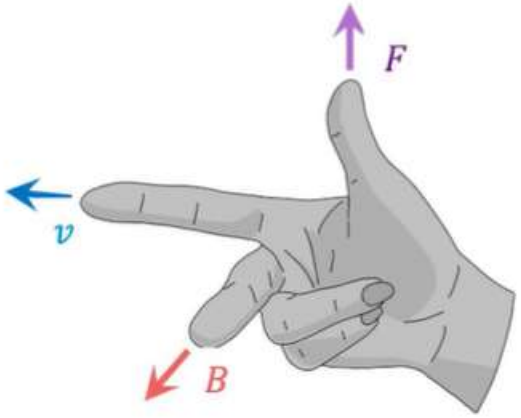
The particle will move in a helical path, centered on the magnetic field.

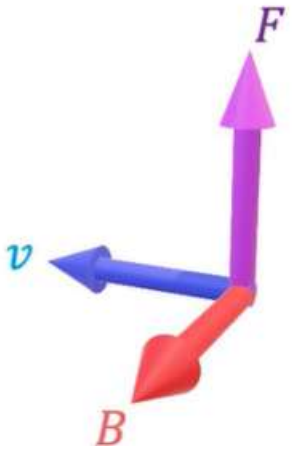
Magnetic Force

The magnetic force acting on a charged object is given by

$$\overrightarrow{F_B} = q\vec{v} \times \vec{B}$$

The direction of the magnetic force acting on a moving charged object can be found using the right-hand rule.





Magnetic Force

The magnetic force acting on a charged object is given by

$$\overrightarrow{F_B} = q\vec{v} \times \vec{B}$$

The magnitude of the force is directly proportional to the magnitude of the **charge**.

If the **charge** doubles, the force doubles.

Magnetic Force

The magnetic force acting on a charged object is given by

$$\overrightarrow{F_B} = q\vec{v} \times \vec{B}$$

The magnitude of the force is directly proportional to the **velocity** of the charged object.

If the **velocity** doubles, the force doubles.

Magnetic Force

The magnetic force acting on a charged object is given by

$$\overrightarrow{F_B} = q\vec{v} \times \vec{B}$$

The magnitude of the force is directly proportional to the strength of the **magnetic field**.

If the strength of the **field** doubles, the force doubles.

Magnetic Force

The magnetic force acting on a charged object is given by

$$\overrightarrow{F_B} = q\vec{v}\times\vec{B}$$

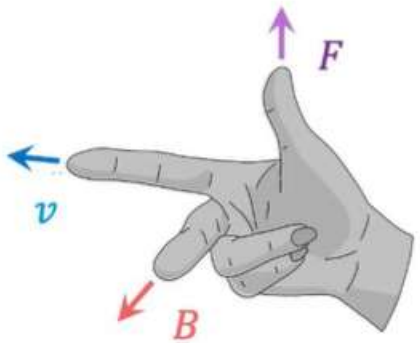
We can find the magnitude of the charge using

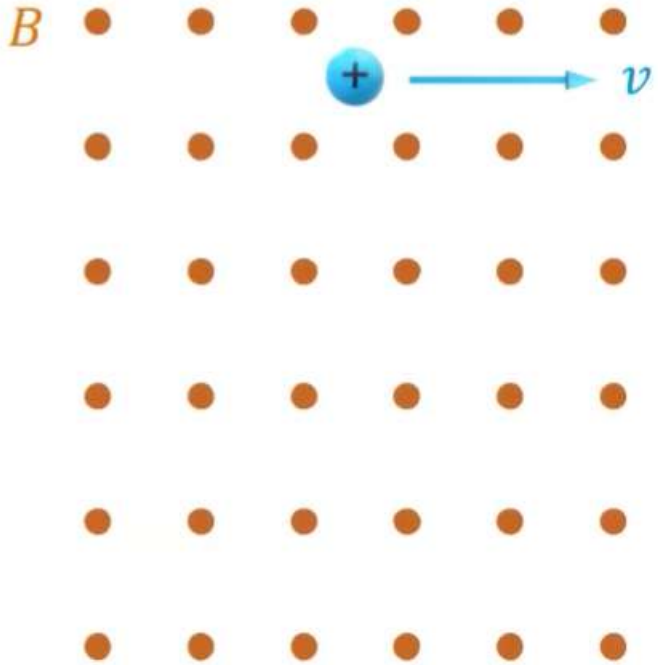
$$F_B = qvB\sin\theta$$

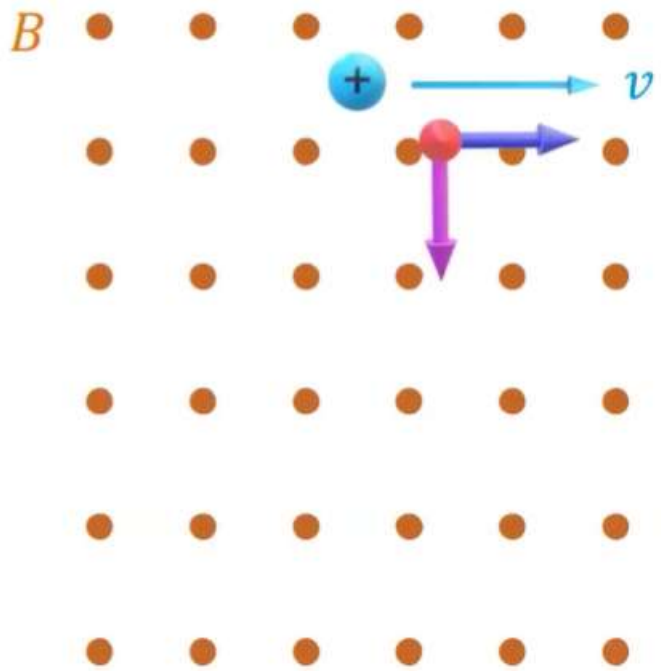
Positively Charged Object Moving in a Field

A positively charged object is moving at a speed of v to the right in a region of space with a uniform magnetic field of magnitude B directed out of the page. What will be the direction of the force on the charged object at the moment shown?

The force on the particle will be perpendicular to both the magnetic field and the velocity, in accordance with the right-hand rule.






B
+
v

Positively Charged Object Moving in a Field

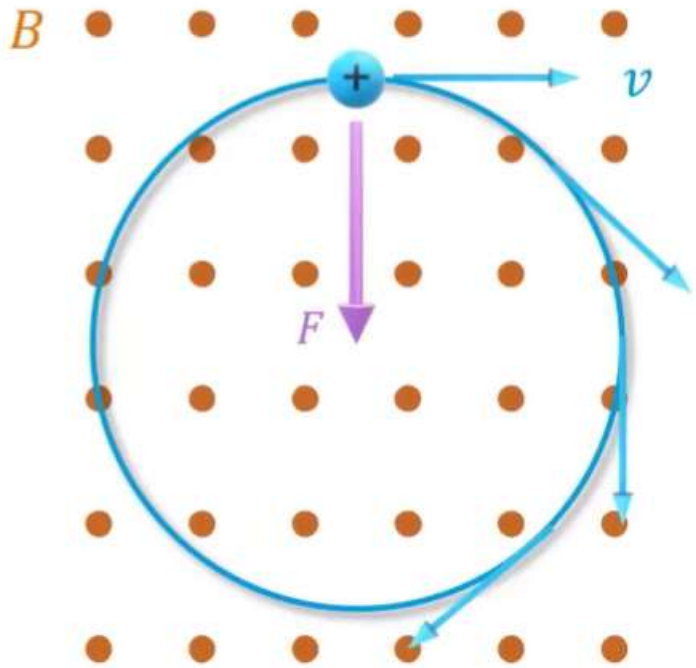
A positively charged object is moving at a speed of v to the right in a region of space with a uniform magnetic field of magnitude B directed out of the page.

Describe the object's motion.

The magnetic force will always be perpendicular to the object's velocity.

The magnetic force acts as a centripetal force, and the object will move in a circle with constant speed.

The magnetic force cannot cause the object to speed up or slow down and does no work on the charged object.



Positively Charged Object Moving in a Field

A positively charged object is moving at a speed of v to the right in a region of space with a uniform magnetic field of magnitude B directed out of the page.

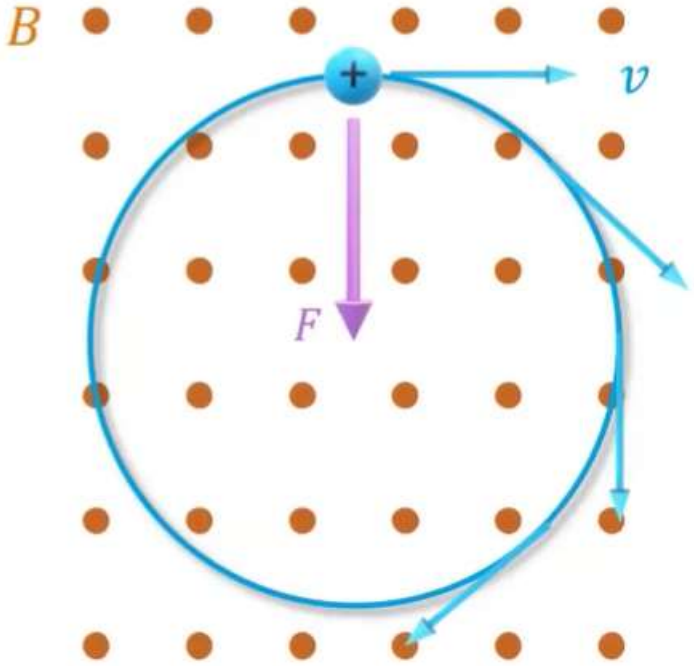
What is the radius of the circle the object moves in?

$$\Sigma F = ma$$

$$F_M = ma_c$$

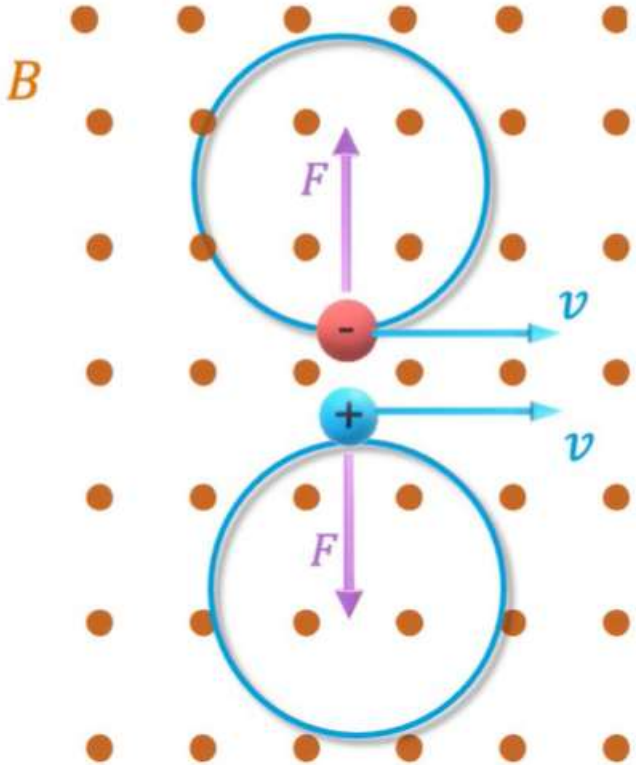
$$qvB = m\frac{v^2}{r}$$

$$r = \frac{mv}{qB}$$



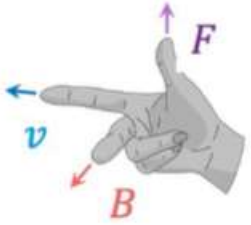
What about a negatively charged object?

The force on a negatively charged object will be directed opposite the force on a positively charged object.

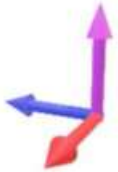


Takeaways

- Magnetic fields exert forces on moving charged particles.
- The direction of the magnetic force on a moving positively charged object can be found with the right-hand rule. The force on a moving negatively charged object is in the opposite direction.



$$\overrightarrow{F_B} = q\vec{v}\times\vec{B}$$



- The magnitude of the magnetic force is defined by: $F_B = qvB\sin\theta$
- The magnetic force is perpendicular to the object's velocity:
  - It can only change the object's direction, not the object's speed.
  - It does no work.

12.2.3

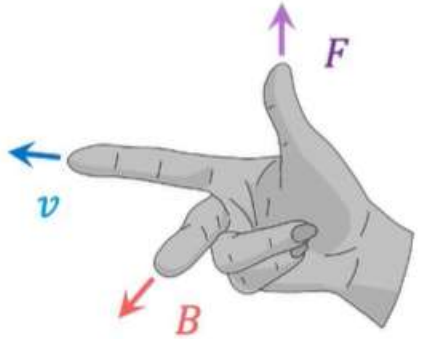
The Motion of a Charged Particle as it moves through

BOTH

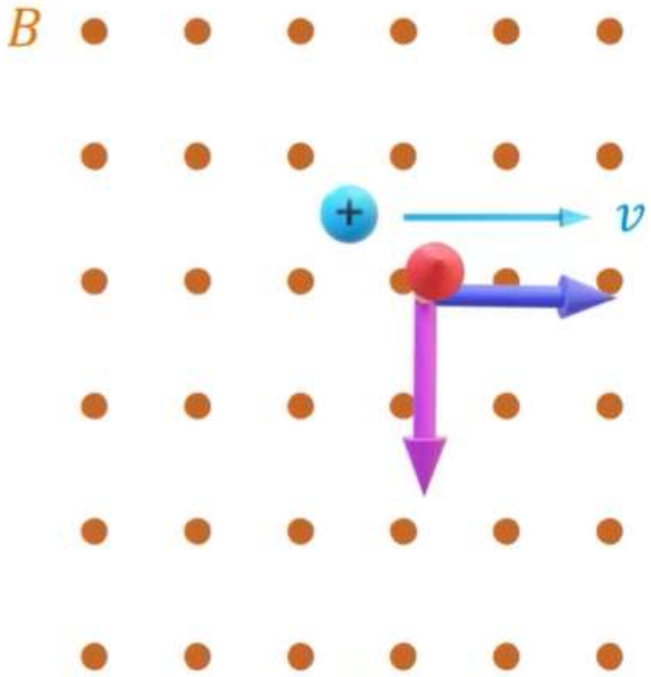
An Electric Field AND a Magnetic Field

Magnetic Force

The force on a positively charged particle moving in a magnetic field will be perpendicular to both the field and the velocity, in accordance with the right-hand rule.

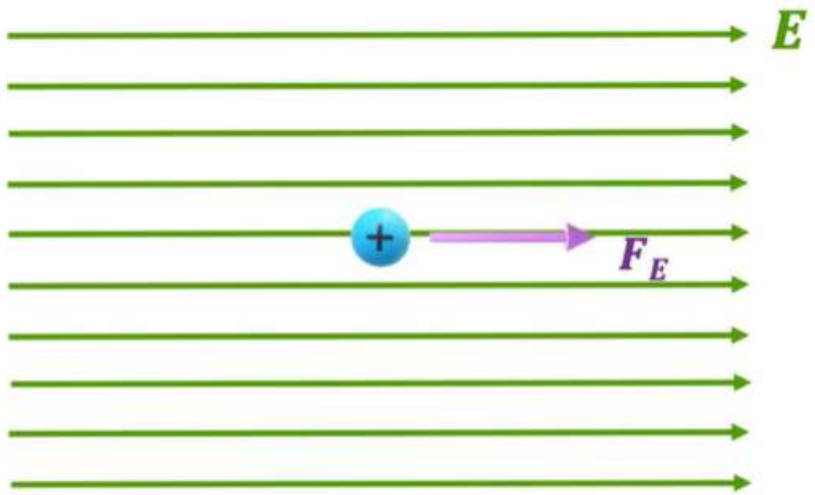


$$F_B = qvB$$



Electric Force

The electric force on a positively charged object will be in the same direction as the electric field.

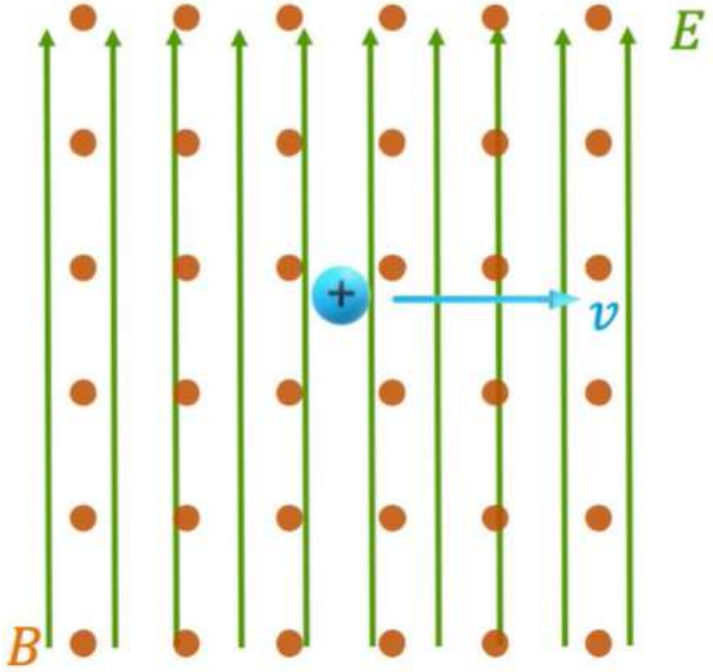


The magnitude of this force is

$$F_E = qE$$

Crossed Fields

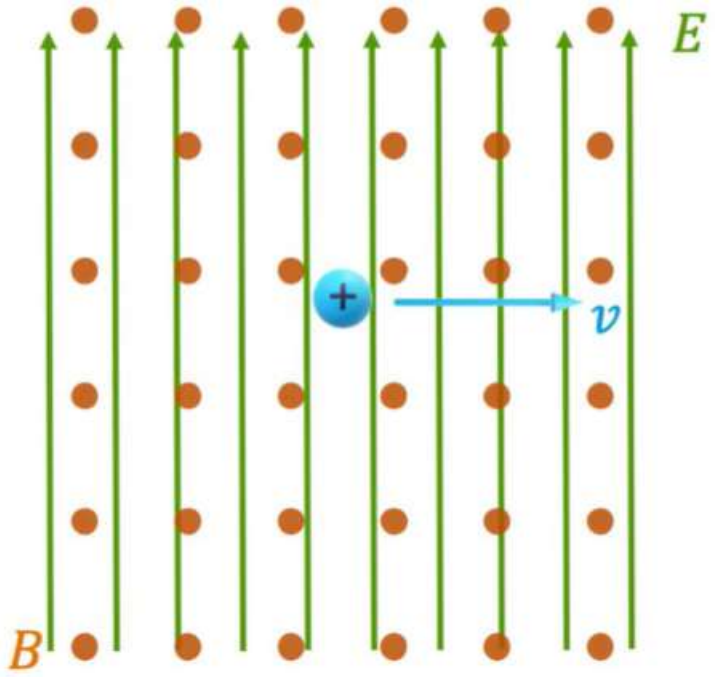
A positively charged particle moves through a region of space that contains both an **electric field** and a **magnetic field**.

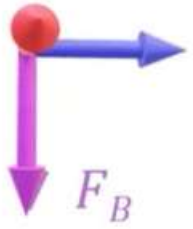


It experiences both a **magnetic** and an **electric force**.

Crossed Fields

Because it's a positively charged particle, the **electric force** acts upward, toward the top of the page.





The **magnetic force** on the particle is downward, toward the bottom of the page.

Crossed Fields

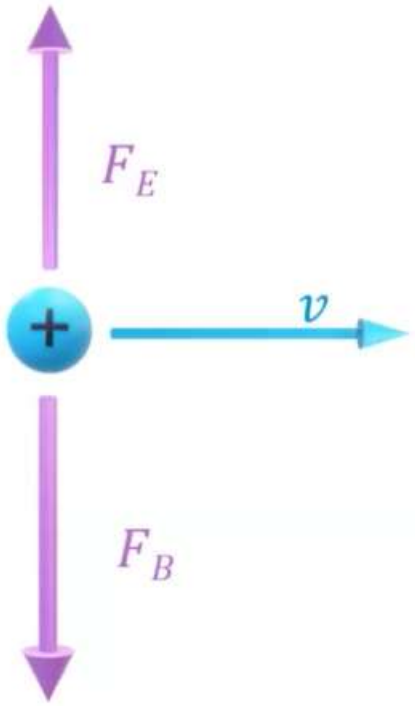
When these forces have the same magnitude and point in opposite directions, the particle will be undeflected and move with a constant velocity.

This velocity can be found by setting the magnitudes of the eclectic and magnetic forces equal to each other.

$$F_E = F_B$$

$$qE = qvB$$

$$v = \frac{E}{B}$$

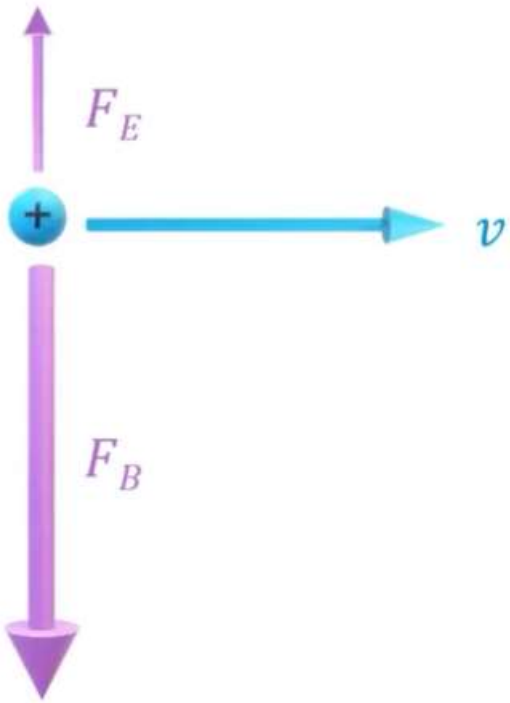


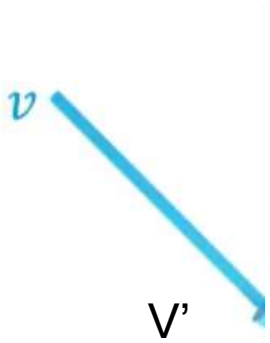
Crossed Fields

$$F_E = F_B$$

$$qE = qvB$$

$$v = \frac{E}{B}$$





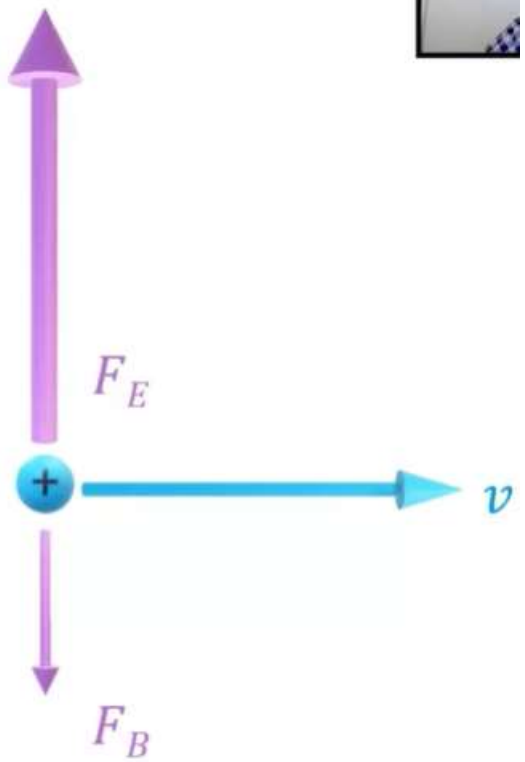
If the particle moves **faster** than this velocity, the magnetic force will be greater than the electric force, and the particle will be deflected downward.

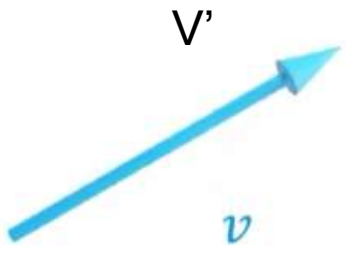
Crossed Fields

$$F_E = F_B$$

$$qE = qvB$$

$$v = \frac{E}{B}$$

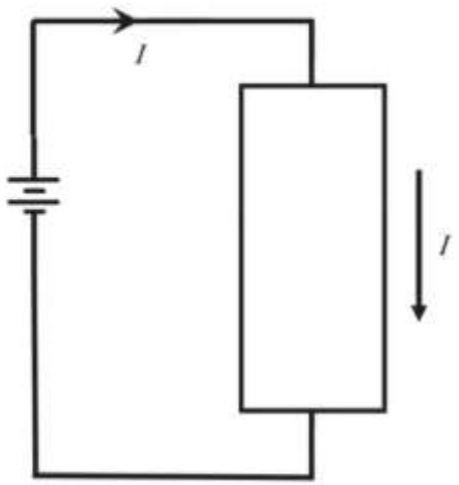


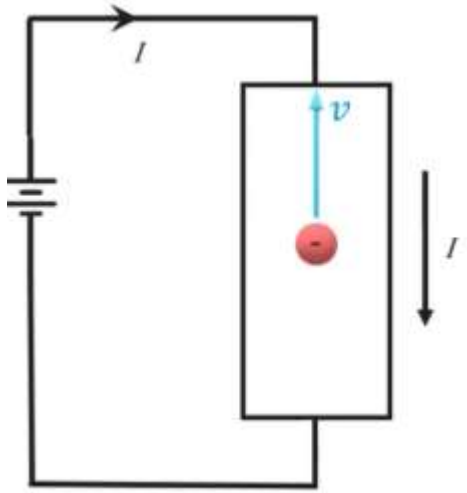


If the particle moves **slower** than this velocity, the electric force will be greater than the magnetic force, and the particle will be deflected upward.

The Hall Effect

Current in the circuit moves clockwise. The current through the flat section of wire shown moves toward the bottom of the page. The electrons in the wire move in the opposite direction of conventional current.



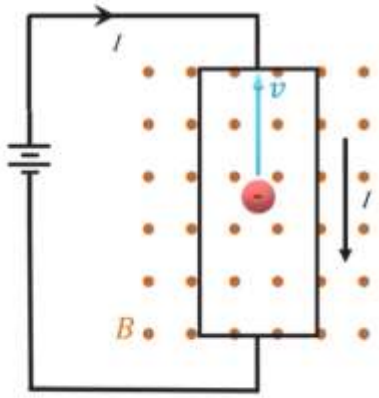


The Hall Effect

Current in the circuit moves clockwise. The current through the flat section of wire shown moves toward the bottom of the page. The electrons in the wire move in the opposite direction of conventional current.

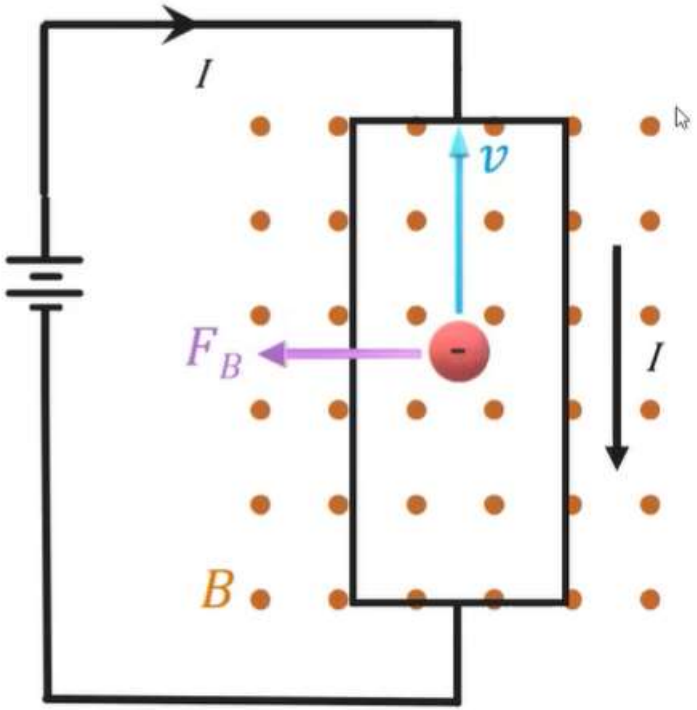
An external **magnetic field** is directed out of the page.

Electrons will be deflected to the left side of the wire by the **magnetic field**.



The Hall Effect

Current in the circuit moves clockwise. The current through the flat section of wire shown moves toward the bottom of the page. The electrons in the wire move in the opposite direction of conventional current.



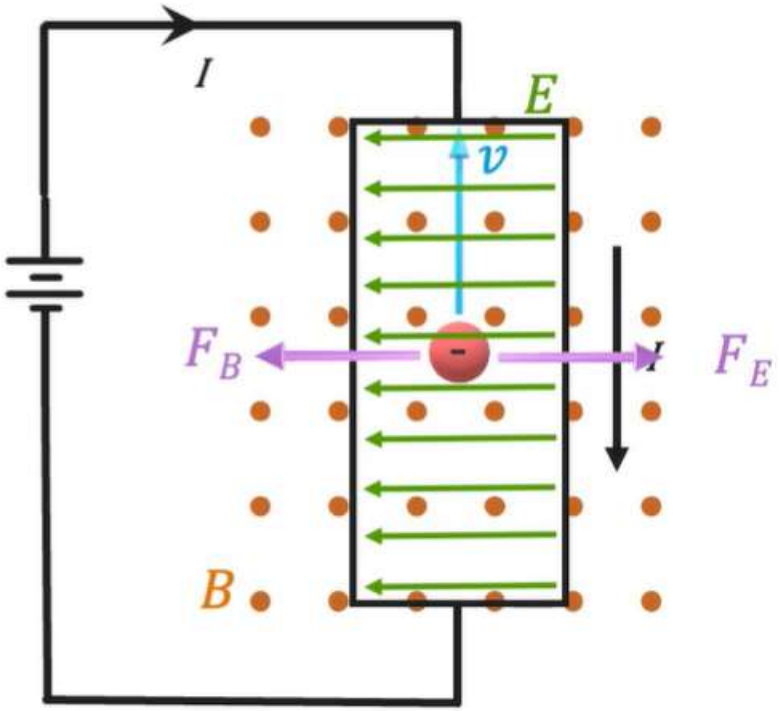
An external **magnetic field** is directed out of the page.

Electrons will be deflected to the left side of the wire by the **magnetic field**.

The Hall Effect

As charge carriers gather on the left side of the wire, they experience a repelling **electric force** that balances the **magnetic force**.

This **electric field** in the wire points to the left as a result of the **magnetic field** being applied.



Takeaways

- In a region containing both a **magnetic** and an **electric** field, a moving charged object will experience independent forces from each field.

- If these forces are balanced on an object, the object will move through the region undeflected, with a **constant velocity**.

- A **magnetic field** applied perpendicularly to a current-carrying wire will produce an electric field in the wire perpendicular to the current (The Hall Effect).